

Smart Energy Report

Generated: 2026-04-27 20:07

AI-Based Smart Energy Monitoring System | INT 428

1. Summary Statistics

Total Consumption:	172066.36 kWh
Avg Daily Usage:	120.07 kWh
Peak Hour:	20:00 - 21:00
Peak Day:	Saturday
Peak Month:	December
Total Records:	157636
Std Deviation:	1.058

2. Trend Analysis

Overall Direction: DECREASING
Avg Daily: 120.07 kWh | Slope: -0.0128 kWh/day

Monthly Breakdown:

Jan:	19745.4 kWh	(- +0.0%)
Feb:	16290.9 kWh	(v -17.5%)
Mar:	16701.6 kWh	(^ +2.5%)
Apr:	13585.1 kWh	(v -18.7%)
May:	14120.7 kWh	(^ +3.9%)
Jun:	11880.1 kWh	(v -15.9%)
Jul:	9587.8 kWh	(v -19.3%)
Aug:	7532.9 kWh	(v -21.4%)
Sep:	12615.5 kWh	(^ +67.5%)
Oct:	15609.1 kWh	(^ +23.7%)
Nov:	16583.8 kWh	(^ +6.2%)
Dec:	17813.3 kWh	(^ +7.4%)

3. AI-Generated Insights

- ? Weekly Consumption Forecast: Your consumption will decrease by 5.0% next week based on the trend from the last 14 days.
- ? Peak vs Off-Peak Ratio: Usage at 20:00 is 4.3× higher than off-peak. Shifting 20% of peak load could save ~13%.
- ? Highest Energy Consumer: AC / Heating accounts for 40.1% of total energy - focus savings efforts here.
- ?? Seasonal Outlook - December: Based on seasonal patterns, December consumption is expected to rise by 7.4%

compared to November.

? Anomaly Health Check: 5565 unusual spikes detected (3.53%). Consider checking appliances for malfunctions.

? Estimated Monthly Cost: At Rs.6.5/kWh, average monthly bill ~ Rs.93,203. Smart scheduling could save 10-15% (Rs.11,184).

? Weekend Usage Pattern: Weekend consumption is 19.1% higher than weekdays - mainly from entertainment and kitchen appliances.

4. Alert Summary

Total Alerts: 182 | Critical: 28 | Warning: 154

Daily Threshold: 190.12 kWh | Hourly Threshold: 3.21 kWh

Days Over Threshold: 102 / 1433

Top Alerts:

[CRITICAL] 2006-12-16 17:37 - Hourly spike 5.27 kWh at 17:37
[CRITICAL] 2006-12-17 - Daily usage 263.3 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2006-12-17 10:44 - Hourly spike 5.87 kWh at 10:44
[CRITICAL] 2006-12-17 10:57 - Hourly spike 4.84 kWh at 10:57
[CRITICAL] 2006-12-17 15:43 - Hourly spike 4.94 kWh at 15:43
[CRITICAL] 2006-12-17 16:22 - Hourly spike 4.84 kWh at 16:22
[CRITICAL] 2006-12-19 08:53 - Hourly spike 5.77 kWh at 08:53
[CRITICAL] 2006-12-19 09:06 - Hourly spike 5.67 kWh at 09:06
[CRITICAL] 2006-12-20 08:04 - Hourly spike 5.97 kWh at 08:04
[CRITICAL] 2006-12-20 18:41 - Hourly spike 4.82 kWh at 18:41
[CRITICAL] 2006-12-23 - Daily usage 348.9 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2006-12-26 - Daily usage 295.3 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2006-12-31 - Daily usage 266.2 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-01-04 - Daily usage 254.1 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-01-21 - Daily usage 267.5 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-02-03 - Daily usage 307.0 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-02-04 - Daily usage 277.1 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-02-11 - Daily usage 287.0 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-02-18 - Daily usage 293.3 kWh exceeded threshold (190.12 kWh)
[CRITICAL] 2007-02-22 - Daily usage 259.9 kWh exceeded threshold (190.12 kWh)

5. Energy-Saving Recommendations

? [HIGH] Reduce Standby Power at Night: Night-time average (0.50 kWh/h) is high. Switch off devices at the plug to save up to 10%. (Save ~10%)

? [MEDIUM] Shift Usage Away from Peak Hour (20:00): Demand peaks at 20:00. Run heavy appliances during off-peak hours to reduce cost. (Save ~8%)

?? [HIGH] Optimise Winter Heating: Winter: 17950 kWh/mo vs mid-season: 13983. Lower thermostat by 1°C to save ~3% on heating. (Save ~6%)

? [MEDIUM] Manage Weekend Consumption: Weekend (1.23 kWh/h) > weekday (1.04). Turn off idle TVs, consoles, and lights. (Save ~5%)

? [HIGH] AC Uses 40% of Total Energy: Consider upgrading to an inverter AC, using smart scheduling, or improving insulation. (Save ~15%)

? [MEDIUM] Switch to Smart LED Lighting: Replace CFL/incandescent with LEDs + motion sensors to cut lighting energy by up to 40%. (Save ~12%)

?? [LOW] Use Energy Star Rated Appliances: 5-star rated fridge/washer can reduce individual appliance consumption

by 20-50%. (Save ~15%)

? [MEDIUM] Install Smart Power Strips: Smart strips cut power to devices on standby automatically, saving 5-8% of phantom load. (Save ~8%)

Report generated by EnergyAI - INT 428 AI Project